

HALL BEACH, NU, Canada

WMO: 713200

Lat: **68.780N**Lon: **81.240W**Elev: **27**StdP: **14.68**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-45.4	-42.5	-49.9	0.3	-43.3	-48.8	0.3	-42.0	32.2	-15.4	29.7	-18.1	10.5	310	1.128

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	57.7	52.0	53.5	48.8	50.3	46.5	52.2	56.9	49.3	53.1	46.8	50.0	9.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
49.1	51.7	55.0	46.3	46.5	51.6	44.0	42.6	48.8	21.6	57.2	19.9	53.2	18.6	50.2	64.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.4	27.8	24.9	-48.2	66.7	4.3	4.0	-51.3	69.6	-53.8	71.9	-56.2	74.2	-59.3	77.1		
			-47.5	57.5	3.6	2.3	-50.1	59.1	-52.1	60.5	-54.2	61.8	-56.7	63.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-23.1	-24.6	-18.1	-1.3	17.9	34.1	43.9	41.9	33.3	19.6	0.1	-13.3
	DBStd	26.53	12.19	11.91	12.19	11.39	9.08	5.28	5.26	4.55	5.00	9.00	12.89	13.34
	HDD50	14840	2265	2090	2111	1540	995	479	202	256	500	943	1498	1961
	HDD65	20295	2730	2510	2576	1990	1460	928	653	716	950	1408	1948	2426
	CDD50	20	0	0	0	0	0	1	14	5	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.9	13.0	13.4	13.5	12.8	12.3	11.4	10.6	11.7	13.5	14.3	14.6	13.4
Precipitation	PrecAvg	8.8	0.3	0.2	0.4	0.5	0.6	0.7	1.1	1.8	1.1	0.9	0.6	0.4
	PrecMax	15.8	0.9	0.9	2.0	1.4	1.9	3.1	3.3	6.3	3.2	2.2	2.0	1.3
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.1	0.1	0.1	0.0	0.0
	PrecStd	2.9	0.2	0.2	0.5	0.4	0.4	0.6	0.7	1.2	0.8	0.5	0.5	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.4	14.3	19.6	30.4	34.7	57.2	65.0	62.6	48.7	34.8	30.1	25.8
		MCWB	21.0	14.0	18.8	29.8	33.8	51.1	55.7	55.2	44.6	33.9	29.4	25.5
	2%	DB	7.8	6.0	11.4	25.3	33.0	48.7	59.4	55.3	44.3	33.1	27.9	18.7
		MCWB	7.1	5.3	10.7	24.6	32.1	45.3	53.0	50.2	41.6	32.4	27.2	18.2
	5%	DB	2.1	0.6	6.8	21.2	31.8	43.9	55.6	51.5	41.7	32.1	24.7	12.6
		MCWB	1.4	-0.1	6.0	20.2	31.1	41.2	50.4	47.6	39.6	31.5	23.9	12.0
	10%	DB	-4.3	-5.5	2.0	17.3	30.4	40.6	52.4	48.7	39.5	30.8	20.9	7.6
		MCWB	-5.0	-6.2	1.3	16.4	29.6	38.7	48.1	45.5	38.1	30.2	20.4	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	14.0	19.0	29.9	34.1	51.4	56.7	55.9	45.1	34.1	29.6	25.6
		MCDB	21.4	14.1	19.5	30.1	34.4	56.2	62.9	61.9	47.9	34.4	29.8	25.7
	2%	WB	7.4	5.5	10.8	24.5	32.4	45.7	53.6	50.9	42.1	32.6	27.3	18.2
		MCDB	8.0	6.2	11.3	25.2	32.8	48.8	58.8	54.4	44.1	32.9	27.8	18.7
	5%	WB	1.5	0.0	6.1	20.2	31.2	41.4	50.7	47.9	39.8	31.6	24.0	12.0
		MCDB	2.1	0.7	6.8	21.0	31.7	43.8	55.1	51.1	41.2	32.0	24.7	12.5
	10%	WB	-4.7	-6.0	1.3	16.4	29.6	38.7	48.3	45.6	38.3	30.3	20.3	7.0
		MCDB	-3.9	-5.4	2.1	17.2	30.3	40.5	52.3	48.3	39.4	30.8	20.9	7.6
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.1	15.3	16.4	12.0	6.7	10.6	8.5	5.8	9.5	13.6	13.1
		MCDBR	17.6	18.9	18.8	15.2	8.4	12.0	16.8	14.0	9.3	6.3	13.1	15.2
		MCWBR	17.6	18.9	18.6	15.0	8.2	9.0	11.7	9.9	7.1	6.5	13.3	15.1
	5% WB	MCDBR	17.5	18.9	18.8	15.1	8.1	11.9	16.7	13.6	8.8	6.2	13.2	15.3
		MCWBR	17.5	18.9	18.6	14.9	8.0	9.1	12.4	10.1	6.9	6.4	13.5	15.2
Clear-Sky Solar Irradiance	taub	0.171	0.204	0.249	0.322	0.323	0.322	0.355	0.326	0.292	0.238	0.159	N/A	
	taud	1.840	2.046	2.158	1.974	2.134	2.303	2.354	2.516	2.516	2.178	1.916	N/A	
	Ebn at Noon	19	189	250	253	272	277	262	260	240	168	21	0	
	Edh at Noon	2	14	24	40	39	34	31	24	18	12	1	0	
All-Sky Solar Radiation	RadAvg	7	150	683	1427	1925	2048	1716	1139	567	163	22	0	
	RadStd	1	15	24	46	114	131	153	75	63	19	3	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page